

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 22

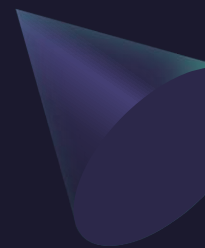


Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.3, 16.4 (fundamental theorem of line integrals and Green's theorem)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Green's Theorem

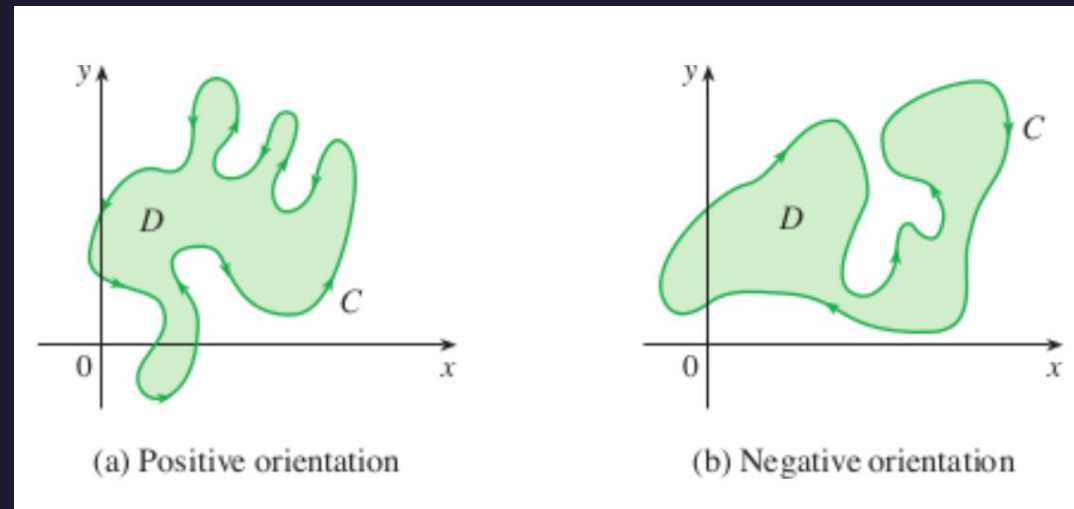


About Parametrized Curves

A parametrized curve is called **closed or a loop** if it starts and ends at the same point.

A closed curve is **simple** if its intersection with itself is at the start and the end, and nowhere else.

A simple closed curve on the plane is **positively oriented** if it goes counterclockwise. Otherwise it is **negatively oriented**.



Green's Theorem

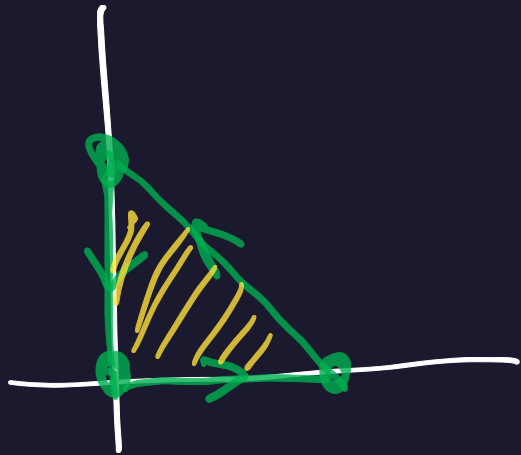
Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$\mathbf{F} = (P, Q)$$

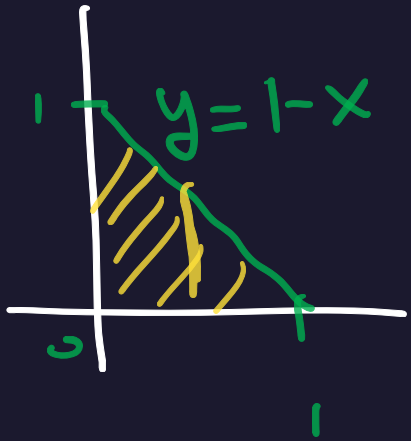
Example: Evaluate $\int_C x^4 dx + xy dy$ where C is the triangular curve along the segments from $(0,0)$ to $(1,0)$, from $(1,0)$ to $(0,1)$, and from $(0,1)$ to $(0,0)$.



$$\int_C x^4 dx + xy dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$\begin{aligned} P &= x^4 & \frac{\partial P}{\partial y} &= 0 \\ Q &= xy & \frac{\partial Q}{\partial x} &= y \end{aligned} \qquad = \iint_D y dA$$

Example: Evaluate $\int_C x^4 dx + xy dy$ where C is the triangular curve along the segments from $(0,0)$ to $(1,0)$, from $(1,0)$ to $(0,1)$, and from $(0,1)$ to $(0,0)$.



$$\iint_D y \, dA$$

$$D = \begin{aligned} 0 &\leq x \leq 1 \\ 0 &\leq y \leq 1-x \end{aligned}$$

$$= \int_0^1 \int_0^{1-x} y \, dy \, dx$$

$$= \int_0^1 \left. \frac{y^2}{2} \right|_0^{1-x} dx = \int_0^1 \frac{(1-x)^2}{2} dx$$

$$= \int_0^1 \frac{1}{2} (1 - 2x + x^2) dx = \boxed{\frac{1}{6}}$$

Green's Theorem

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\oint_C P dx + Q dy = \int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

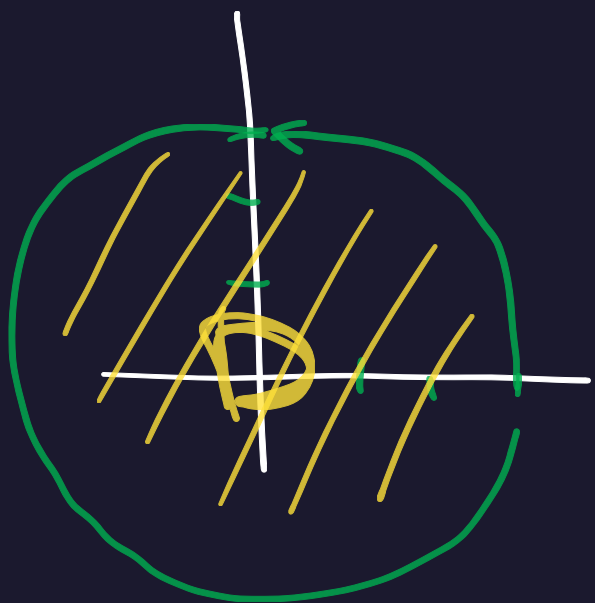
Notes: Sometimes we write $\oint_C$... to denote we are integrating along a closed curve.

If $F = (P, Q)$ is conservative, then $F = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$ and therefore

$$\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x}$$

$$\iint_D \underbrace{\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}}_{=0} dA = 0$$

Example: Evaluate $\oint_C (3y - e^{\sin(x)})dx + (7x + \sqrt{y^4 + 1})dy$
where C is the circle of radius 3 centered at the origin.



$$r(t) = (3\cos t, 3\sin t)$$
$$0 \leq t \leq 2\pi$$

$$P = 3y - e^{\sin x}$$

$$\frac{\partial P}{\partial y} = 3$$

$$Q = 7x + \sqrt{y^4 + 1}$$

$$\frac{\partial Q}{\partial x} = 7$$

$$\left. \begin{array}{l} \frac{\partial P}{\partial y} = 3 \\ \frac{\partial Q}{\partial x} = 7 \end{array} \right\} \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

$$= 7 - 3 = 4$$

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_D 4 dA$$

Example: Evaluate $\oint_C (3y - e^{\sin(x)})dx + (7x + \sqrt{y^4 + 1})dy$
where C is the circle of radius 3 centered at the origin.

$$\iint_D 4 \, dA = 4 \cdot \underbrace{\iint_D 1 \cdot dA}_{\int_0^{2\pi} \int_0^3 r \, dr \, d\theta} = 4 \cdot (\pi \cdot 3^2) = \boxed{36\pi}$$

↑
circle
of radius 3

Green's Theorem

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$$

$$\int_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$P(x, y) = 0$	$P(x, y) = -y$	$P(x, y) = -\frac{1}{2}y$
$Q(x, y) = x$	$Q(x, y) = 0$	$Q(x, y) = \frac{1}{2}x$

$$A = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C x \, dy - y \, dx$$

Example: Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

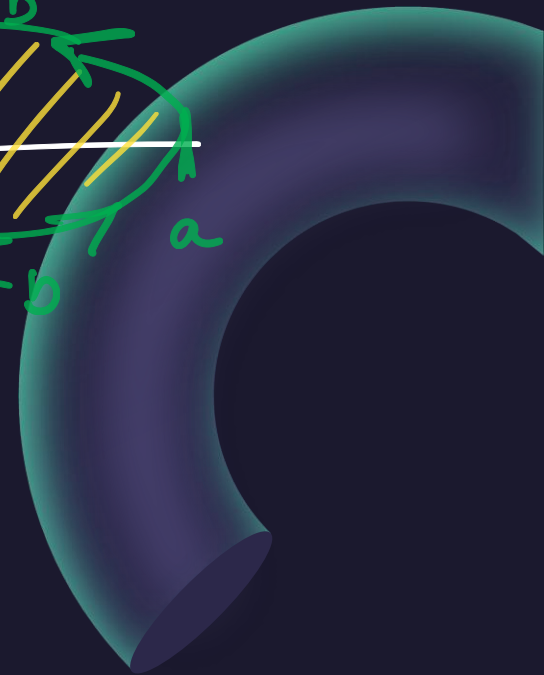
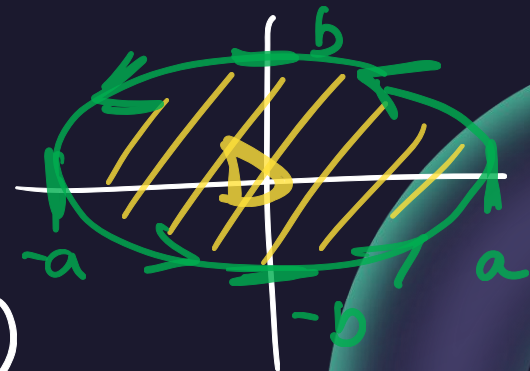
$$A = \iint_D 1 \, dA \stackrel{\text{GT}}{=} \frac{1}{2} \oint_C (x \, dy - y \, dx)$$

$$C = \begin{cases} x = a \cos t \\ y = b \sin t \end{cases} \quad 0 \leq t \leq 2\pi$$

$$dx = -a \sin t \, dt$$

$$dy = b \cos t \, dt$$

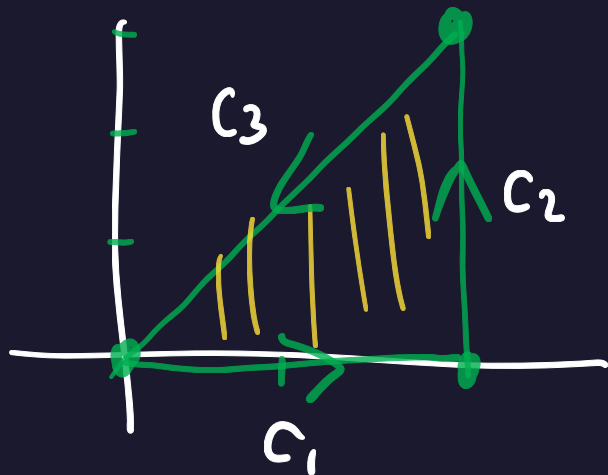
$$\begin{aligned} x \, dy - y \, dx &= a \cos t \cdot b \cos t \, dt - b \sin t (-a \sin t \, dt) \\ &= ab \cdot (\cos^2 t + \sin^2 t) \, dt = ab \cdot dt \end{aligned}$$



Example: Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

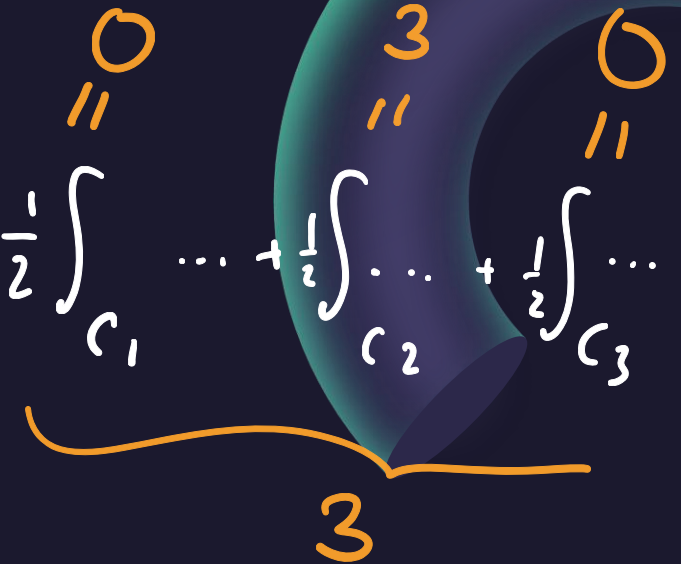
$$\begin{aligned} A &= \iint_D 1 \, dA = \frac{1}{2} \oint_C (x \, dy - y \, dx) \\ &= \frac{1}{2} \int_0^{2\pi} (ab) \, dt = \frac{ab}{2} \int_0^{2\pi} 1 \, dt = \frac{ab}{2} \cdot 2\pi \\ &= \boxed{ab\pi} \end{aligned}$$

Example: Find the area of the triangle with vertices $(0,0)$, $(2,0)$, $(2,3)$.



$$\text{Area} = \frac{2 \cdot 3}{2} = 3$$

$$= \frac{1}{2} \oint_C (x dy - y dx) = \frac{1}{2} \int_{C_1} \dots + \frac{1}{2} \int_{C_2} \dots + \frac{1}{2} \int_{C_3} \dots$$

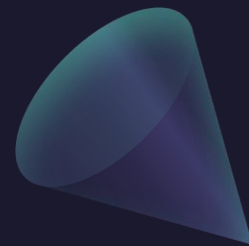


$$C_1 : \begin{cases} x = t & 0 \leq t \leq 2 \\ y = 0 \end{cases} \quad \left. \begin{array}{l} dx = dt \\ dy = 0 \cdot dt \end{array} \right\}$$

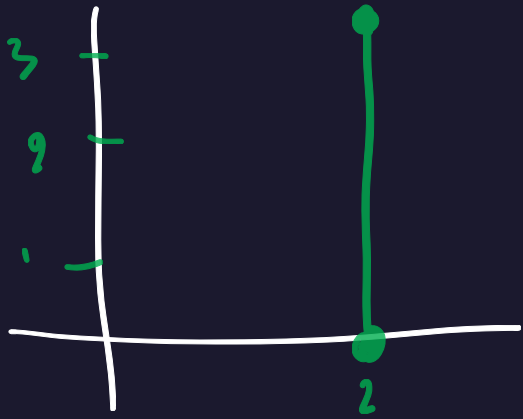
$$C_2 : \begin{cases} x = 2t & 0 \leq t \leq 1 \\ y = 0 \end{cases} \quad x dy - y dx = t \cdot 0 dt - 0 \cdot dt = 0 \cdot dt$$

$$C_3 : \begin{cases} x = 2t \\ y = 0 \end{cases}$$

$$\frac{1}{2} \int_0^2 0 dt = 0$$



Example: Find the area of the triangle with vertices $(0,0)$, $(2,0)$, $(2,3)$.



$$\frac{1}{2} \int_{C_2} x dy - y dx = \frac{1}{2} \int_0^3 2 dt$$

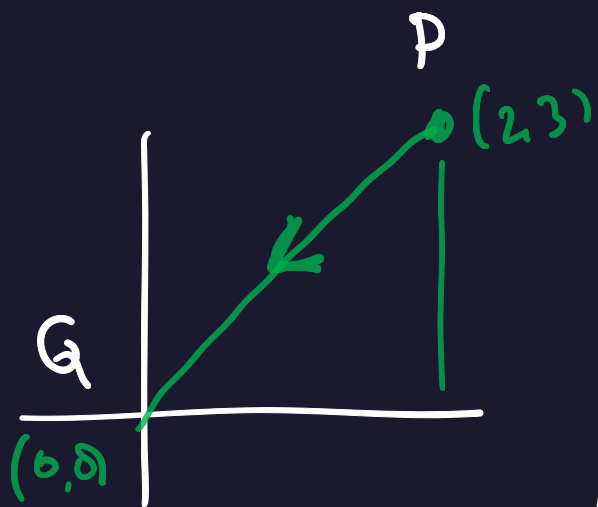
$$= \int_0^3 dt = t \Big|_0^3 = 3$$

$$C_2 = \begin{cases} x = 2 & dx = 0 \cdot dt \\ y = t & dy = dt \end{cases}$$

$$0 \leq t \leq 3$$

$$\begin{aligned} x dy - y dx &= 2 dt - t \cdot 0 dt \\ &= 2 dt \end{aligned}$$

Questions?



$$C_3: \begin{cases} x = 2 - 2t & dx = -2dt \\ y = 3 - 3t & dy = -3dt \end{cases}$$

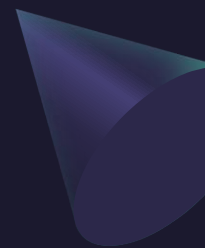
$$0 \leq t \leq 1 \quad x dy - y dx = (2 - 2t)(-3dt) - (3 - 3t)(-2dt)$$

$$r(t) = (1-t)P + tQ = 0 dt$$

$$\frac{1}{2} \oint_{C_3} x dy - y dx = \frac{1}{2} \int_0^1 0 dt = 0$$



ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Curl and Divergence



Curl

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and the partial derivatives of P , Q , and R all exist, then the curl of $\mathbf{F}$ is the vector field on $\mathbb{R}^3$ defined by

1

$$\nabla(f) = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \quad \frac{d}{dx}(f)$$

$$\text{curl } \mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$$

$$\begin{aligned} \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \quad - \left(\frac{\partial}{\partial x} R - \frac{\partial}{\partial z} P \right) \\ &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \\ &= \text{curl } \mathbf{F} \end{aligned}$$

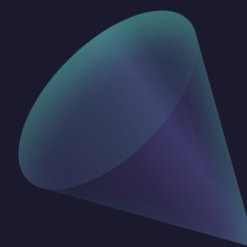
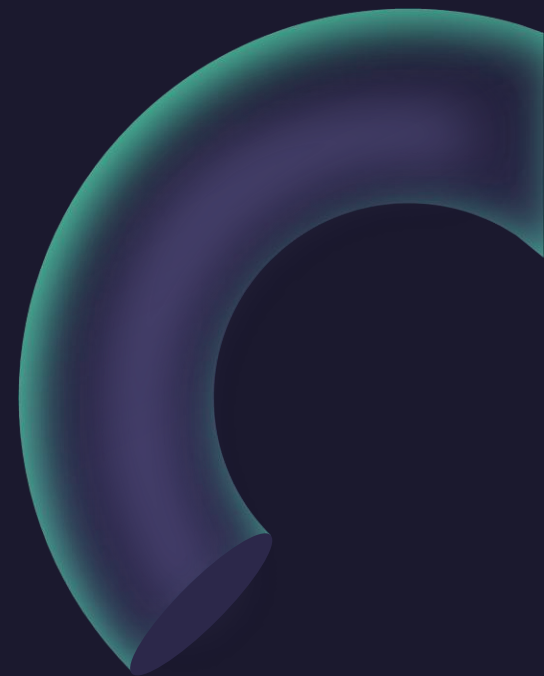
Example: Find the curl of $F(x, y, z) = (xz, xyz, -y^2)$.

$$\text{Curl } F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xyz & -y^2 \end{vmatrix}$$

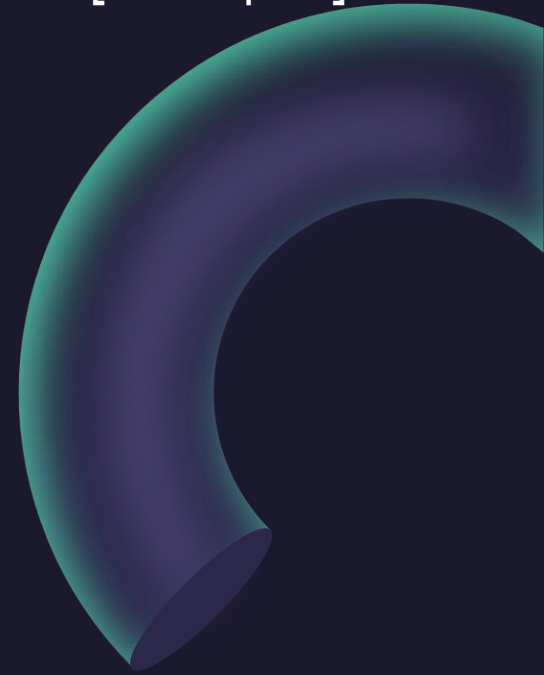
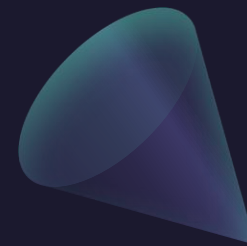
$$= (-2y - xy) \mathbf{i} - (0 - x) \mathbf{j} + (yz - 0) \mathbf{k}$$

$$= (-2y - xy) \mathbf{i} + x \mathbf{j} + yz \mathbf{k}$$

$$= (-2y - xy, x, yz)$$



Example: Find the curl of $F(x, y, z) = (xz, xyz, -y^2)$.



Curl

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and the partial derivatives of P , Q , and R all exist, then the curl of $\mathbf{F}$ is the vector field on $\mathbb{R}^3$ defined by

1

$$\text{curl } \mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

3 Theorem

If f is a function of three variables that has continuous second-order partial derivatives, then

$$\text{curl } (\nabla f) = \mathbf{0}$$

Curl

3 Theorem

If f is a function of three variables that has continuous second-order partial derivatives, then

$$\operatorname{curl}(\nabla f) = \mathbf{0}$$

Proof

We have

$$\begin{aligned}\operatorname{curl}(\nabla f) &= \nabla \times (\nabla f) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} \\ &= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \mathbf{i} + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \right) \mathbf{j} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) \mathbf{k} \\ &= 0 \mathbf{i} + 0 \mathbf{j} + 0 \mathbf{k} = \mathbf{0}\end{aligned}$$

by Clairaut's Theorem.

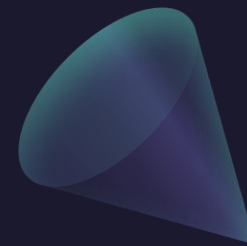
Example: Show that the vector field $F(x, y, z) = (xz, xyz, -y^2)$ is NOT conservative.

$$\iff F = \nabla f \Rightarrow \text{curl } F = \text{curl}(\nabla f) = 0$$

$$\text{Curl}(F) = (-2y - xy, x, yz) \neq (0, 0, 0)$$

$$\Rightarrow F \text{ is NOT } \underline{\text{conservative}},$$

Example: Show that the vector field $F(x, y, z) = (xz, xyz, -y^2)$ is NOT conservative.



Curl

3 Theorem

If f is a function of three variables that has continuous second-order partial derivatives, then

$$\operatorname{curl}(\nabla f) = \mathbf{0}$$

4 Theorem

If $\mathbf{F}$ is a vector field defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives and $\operatorname{curl} \mathbf{F} = \mathbf{0}$, then $\mathbf{F}$ is a conservative vector field.

Example: Show that the vector field

$F(x, y, z) = (y^2 z^3, 2xyz^3, 3xy^2 z^2)$ is conservative.

$$\text{Curl}(F) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 z^3 & 2xyz^3 & 3xy^2 z^2 \end{vmatrix}$$

$$= \underbrace{\left(6xyz^2 - 6xyz^2 \right)}_0 \mathbf{i} - \underbrace{\left(3y^2 z^2 - 3y^2 z^2 \right)}_0 \mathbf{j} + \underbrace{\left(2yz^3 - 2yz^3 \right)}_0 \mathbf{k}$$
$$= (0, 0, 0) \Rightarrow \underline{\text{conservative!}}$$

Example: Find a function $f(x,y,z)$ such that its gradient is given by

$$\nabla f = F(x,y,z) = (y^2 z^3, 2xyz^3, 3xy^2 z^2).$$

$$\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \frac{\partial f}{\partial z}$$

$$f = \int y^2 z^3 dx = \boxed{xy^2 z^3} + G(y,z)$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (xy^2 z^3 + C(y,z)) = 2xyz^3 + \frac{\partial}{\partial y} C(y,z)$$

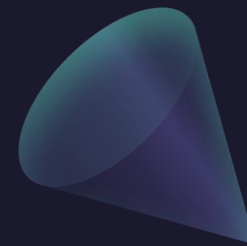
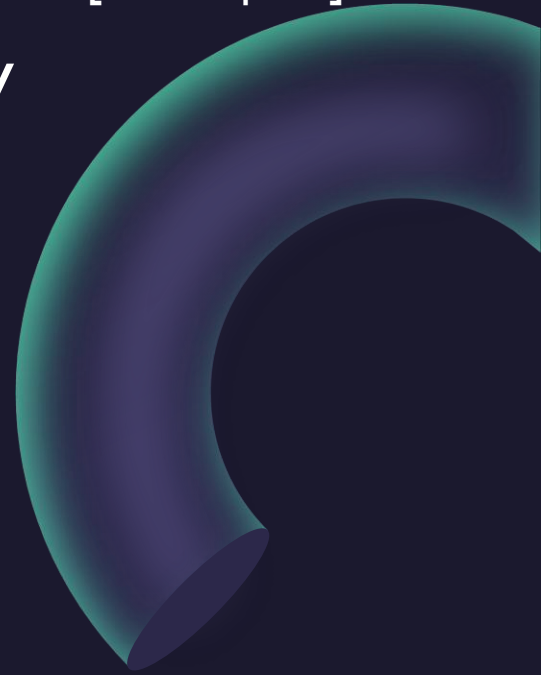
$$\Rightarrow C(y,z) = C(z)$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z} (xy^2 z^3 + C(z)) = 3xy^2 z^2 + C'(z)$$

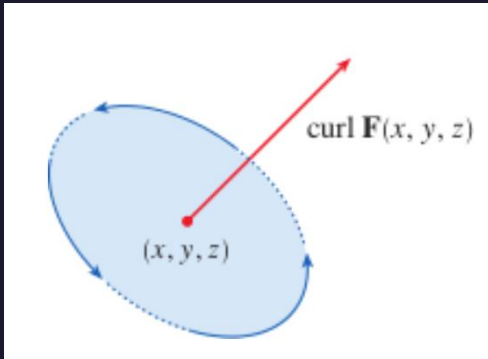
$$= 3xy^2 z^2 + 0$$

$$\boxed{f = xy^2 z^3 \Rightarrow \nabla f = \vec{F}}$$

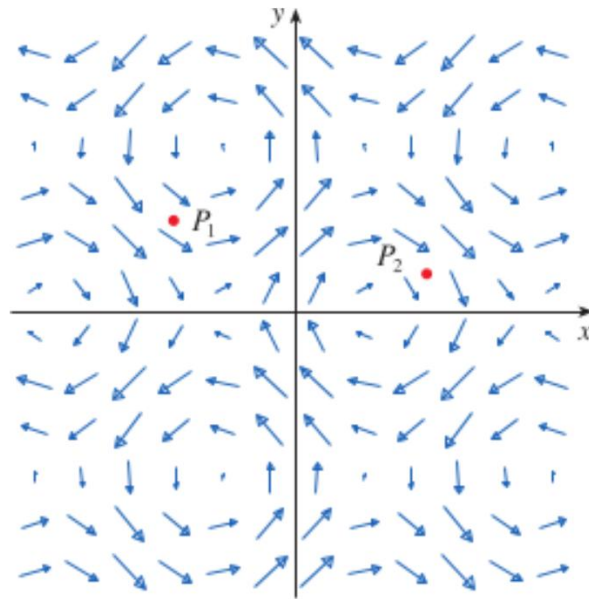
Example: Find a function $f(x,y,z)$ such that its gradient is given by
$$F(x,y,z) = (y^2z^3, 2xyz^3, 3xy^2z^2).$$



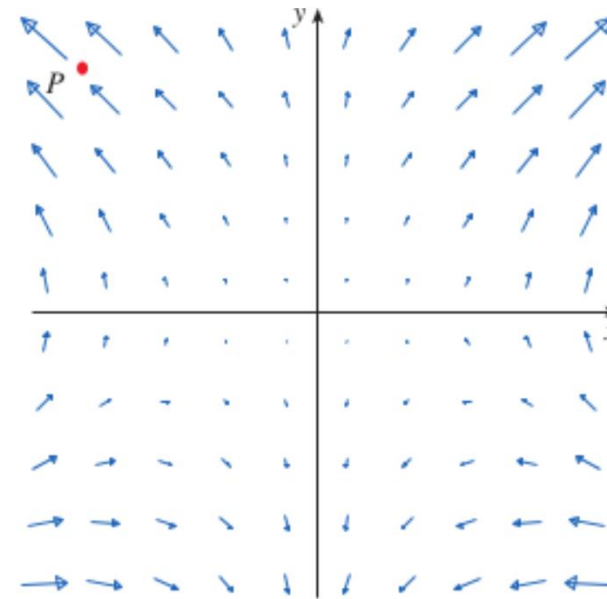
Curl



Velocity fields in fluid flow. (Only the part of $\mathbf{F}$ in the xy -plane is shown; the vector field looks the same in all horizontal planes because $\mathbf{F}$ is independent of z and the z -component is 0.)



(a) $\mathbf{F}(x, y, z) = \sin y \mathbf{i} + \cos x \mathbf{j}$
 $\text{curl } \mathbf{F}(x, y, z) = -(\sin x + \cos y) \mathbf{k}$



(b) $\mathbf{F}(x, y, z) = 2xy \mathbf{i} + (x^2 + y) \mathbf{j}$
 $\text{curl } \mathbf{F}(x, y, z) = \mathbf{0}$

Divergence

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and $\partial P/\partial x$, $\partial Q/\partial y$ and $\partial R/\partial z$ exist, then the **divergence of $\mathbf{F}$** is the function of three variables defined by

9

$$\operatorname{div} \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Divergence

$$\text{CURL}(\mathbf{F}) = \nabla \times \mathbf{F}$$

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and $\partial P/\partial x$, $\partial Q/\partial y$ and $\partial R/\partial z$ exist, then the **divergence of $\mathbf{F}$** is the function of three variables defined by

9

$$\text{div } \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

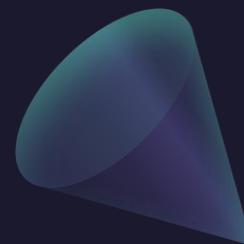
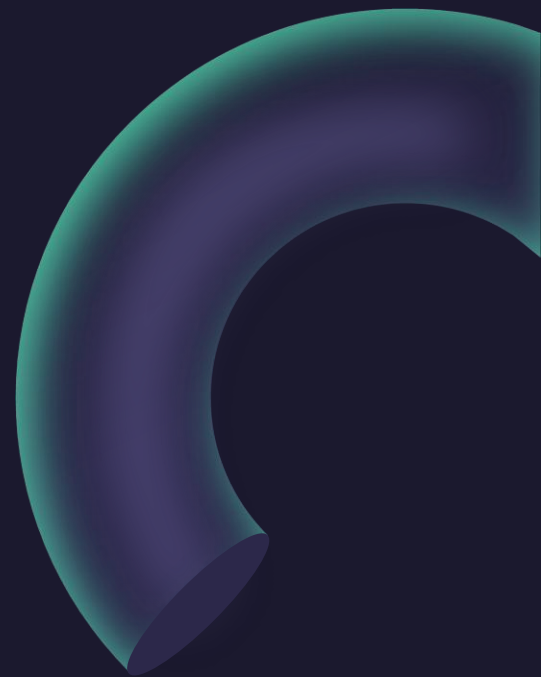
DOT
product

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

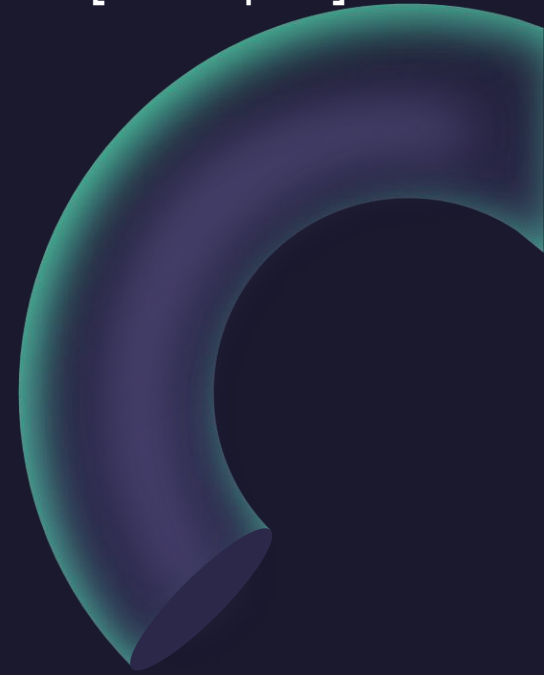
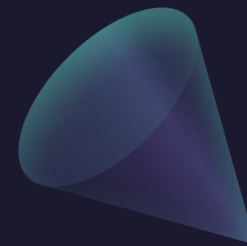
$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$$

Example: Find the divergence of $F(x, y, z) = (xz, xyz, -y^2)$.

$$\begin{aligned}\operatorname{div} F &= \nabla \cdot F \\ &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (xz, xyz, -y^2) \\ &= (z, xz, 0)\end{aligned}$$



Example: Find the divergence of $F(x, y, z) = (xz, xyz, -y^2)$.



Divergence

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and $\partial P/\partial x$, $\partial Q/\partial y$ and $\partial R/\partial z$ exist, then the **divergence of $\mathbf{F}$** is the function of three variables defined by

9

$$\operatorname{div} \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \qquad \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$$

11 Theorem

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and P , Q , and R have continuous second-order partial derivatives, then

$$\operatorname{div} \operatorname{curl} \mathbf{F} = 0$$

Divergence

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$$

11 Theorem

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and P , Q , and R have continuous second-order partial derivatives, then

$$\operatorname{div} \operatorname{curl} \mathbf{F} = 0$$

Proof

Using the definitions of divergence and curl, we have

$$\begin{aligned}\operatorname{div} \operatorname{curl} \mathbf{F} &= \nabla \cdot (\nabla \times \mathbf{F}) \\&= \frac{\partial}{\partial x} \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \\&= \frac{\partial^2 R}{\partial x \partial y} - \frac{\partial^2 Q}{\partial x \partial z} + \frac{\partial^2 P}{\partial y \partial z} - \frac{\partial^2 R}{\partial y \partial x} + \frac{\partial^2 Q}{\partial z \partial x} - \frac{\partial^2 P}{\partial z \partial y} \\&= 0\end{aligned}$$

because the terms cancel in pairs by Clairaut's Theorem.

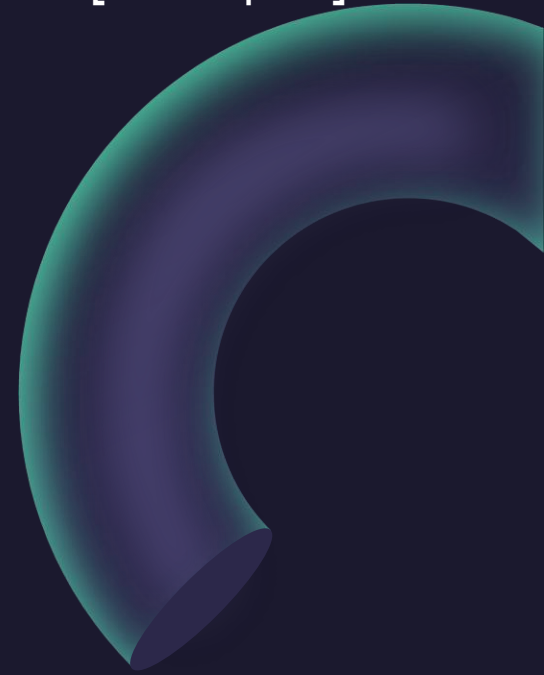
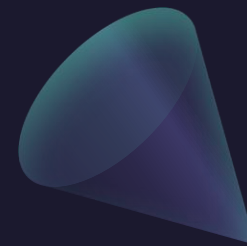
Example: Show that the vector field $F(x, y, z) = (xz, xyz, -y^2)$ is not a curl, i.e., there is no vector field G such that $F = \text{curl } G$.

If $F = \text{Curl } G$, then $\text{div } F = \text{div } \text{Curl } G = 0$

Just compute $\text{div } F = (x, xz, 0) \neq 0$

$\Rightarrow F$ is NOT a Curl.

Example: Find the divergence of $F(x, y, z) = (xz, xyz, -y^2)$.

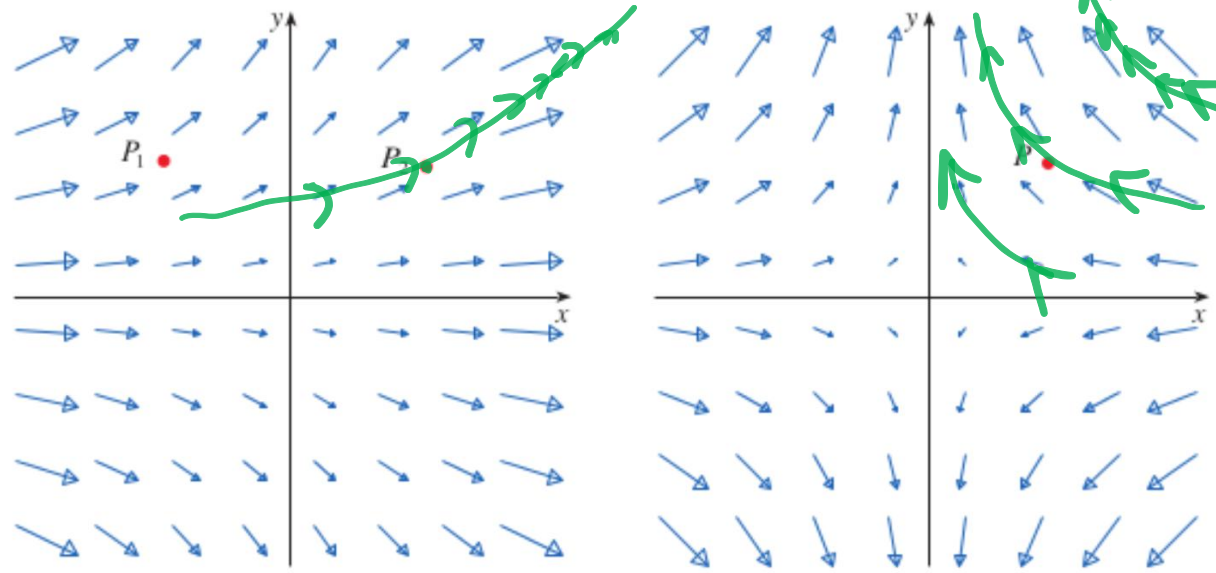


Divergence

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$$

Again, the reason for the name *divergence* can be understood in the context of fluid flow. If $\mathbf{F}(x, y, z)$ is the velocity of a fluid (or gas), then $\operatorname{div} \mathbf{F}(x, y, z)$ represents the net rate of change (with respect to time) of the mass of fluid (or gas) flowing from the point (x, y, z) per unit volume. In other words, $\operatorname{div} \mathbf{F}(x, y, z)$ measures the tendency of the fluid to diverge from the point (x, y, z) . If $\operatorname{div} \mathbf{F} = 0$, then $\mathbf{F}$ is said to be **incompressible**.

Velocity fields in fluid flow. (Only the part of $\mathbf{F}$ in the xy -plane is shown; the vector field looks the same in all horizontal planes because $\mathbf{F}$ is independent of z and the z -component is 0.)



(a) $\mathbf{F}(x, y, z) = (1 + x^2)\mathbf{i} + y\mathbf{j}$
 $\operatorname{div} \mathbf{F}(x, y, z) = 2x + 1$

(b) $\mathbf{F}(x, y, z) = -x\mathbf{i} + y\mathbf{j}$
 $\operatorname{div} \mathbf{F}(x, y, z) = 0$

Laplace

Another differential operator occurs when we compute the divergence of a gradient vector field ∇f .

If f is a function of three variables, we have

$$\operatorname{div}(\nabla f) = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

↑
DOT

and this expression occurs so often that we abbreviate it as $\nabla^2 f$. The operator

$$\nabla^2 = \nabla \cdot \nabla$$

is called the **Laplace operator** because of its relation to **Laplace's equation**

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

$$\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

$$\nabla^2 \mathbf{F} = \nabla^2 P \mathbf{i} + \nabla^2 Q \mathbf{j} + \nabla^2 R \mathbf{k}$$

Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P \, dx + Q \, dy$$

Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P \, dx + Q \, dy$$

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P(x, y) & Q(x, y) & 0 \end{vmatrix} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P \, dx + Q \, dy$$

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P(x, y) & Q(x, y) & 0 \end{vmatrix} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

Therefore:

$$(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \cdot \mathbf{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P \, dx + Q \, dy$$

$$(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \cdot \mathbf{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Then Green's Theorem says:

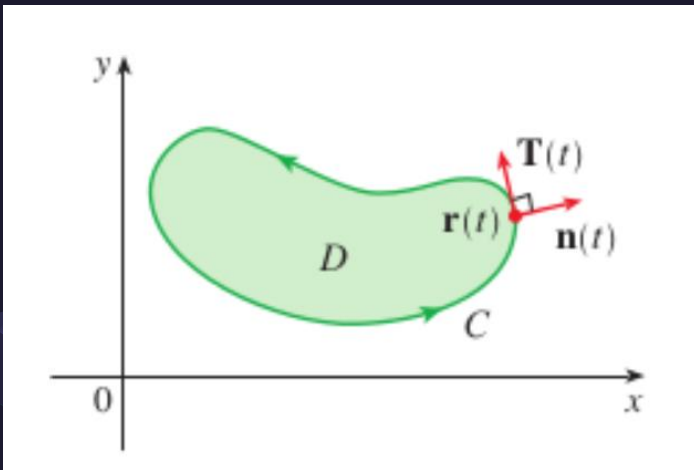
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_D (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dA$$

Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P dx + Q dy$$

Then Green's Theorem alternatively says:



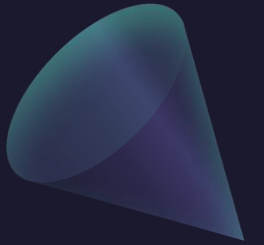
$$\mathbf{r}(t) = x(t) \mathbf{i} + y(t) \mathbf{j} \quad a \leq t \leq b$$

$$\mathbf{T}(t) = \frac{x'(t)}{|\mathbf{r}'(t)|} \mathbf{i} + \frac{y'(t)}{|\mathbf{r}'(t)|} \mathbf{j}$$

$$\mathbf{n}(t) = \frac{y'(t)}{|\mathbf{r}'(t)|} \mathbf{i} - \frac{x'(t)}{|\mathbf{r}'(t)|} \mathbf{j}$$

$$\oint_C \mathbf{F} \cdot \mathbf{n} ds = \iint_D \operatorname{div} \mathbf{F}(x, y) dA$$

Questions?



Thank you

Until next time.

